

RAINFALL at GaretonCounty of South LancashireObserver J. A. Brodie Esq. Ch.River Basin Lower Mersey 2H. Owens Esq. Ch.F. W. Borden 1919☐ Rain Gauge, by Ipwell 1912

| Diameter ... | 10   |       |       |       |       |       |       |       |       |       |        |
|--------------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|
| Height ..... | 110  |       |       |       |       |       |       |       |       |       |        |
| Altitude ... | 847  |       |       |       |       |       |       |       |       |       |        |
| Year ...     | 1910 | 1911  | 1912  | 1913  | 1914  | 1915  | 1916  | 1917  | 1918  | 1919  | MEANS. |
| January ...  |      | .73   | 4.63  | 3.03  | 1.60  | 4.20  | 1.64  | 2.38  | 2.16  | 3.82  |        |
| February ... |      | 2.52  | 1.28  | 1.14  | 1.92  | 2.97  | 4.10  | 2.00  | 2.85  | 1.84  |        |
| March .....  |      | 1.34  | 5.13  | 3.10  | 3.42  | 2.21  | 2.14  | 3.13  | 1.38  | 7.13  |        |
| April .....  |      | 2.26  | .22   | 3.91  | .95   | 1.95  | 2.26  | 2.25  | 1.62  | 1.92  |        |
| May .....    |      | .64   | 2.23  | 3.23  | 2.72  | 1.96  | 3.40  | 3.85  | 3.71  | 1.10  |        |
| June .....   |      | 3.54  | 3.59  | 2.38  | 2.43  | 1.08  | 2.67  | 2.90  | 2.63  | 2.31  |        |
| July .....   |      | .81   | 3.03  | 2.68  | 5.54  | 4.20  | 1.76  | 1.98  | 5.82  | 1.96  |        |
| August ..... |      | 3.23  | 7.06  | 2.67  | 2.81  | 4.82  | 3.04  | 7.80  | 2.54  | 5.13  |        |
| September .. |      | 4.78  | 1.41  | 1.83  | 3.28  | .96   | 3.22  | 1.47  | 8.11  | 2.90  |        |
| October ...  |      | 3.44  | 3.12  | 3.04  | 2.25  | 2.46  | 4.92  | 7.73  | 2.77  | 3.90  |        |
| November ..  |      | 2.54  | 3.04  | 3.31  | 4.03  | 2.71  | 2.99  | 4.12  | 2.87  | 2.44  |        |
| December ... |      | 4.25  | 2.71  | 3.09  | 5.71  | 6.18  | 2.93  | 2.14  | 7.61  | 6.67  |        |
| TOTALS ...   |      | 30.12 | 37.45 | 33.41 | 36.66 | 35.70 | 35.07 | 41.75 | 44.07 | 41.12 |        |

1912. Iron Rainfall 5'6" high 6 ft. enclosing S.

Record 1911-1929 generally too big

A new gauge was installed on a new site in 1930.

Annual values rejected from surveys. (6" high)

| Date.    | Lat. N.   | Long. | Railway Station.  | Distance. | Direction. | Church. | Distance. | Direction. |
|----------|-----------|-------|-------------------|-----------|------------|---------|-----------|------------|
| 53.21.35 | 2.54.35 W |       | Cresanybriarsdale | 1200 ft   | S          | St Mary | 840 ft    | S          |

RAINFALL at GarsingtonCounty of LancashireObserver J. A. Brodie Esq. C.E.River Basin Lower Mersey 2D. P. Rank, C.E.  
City Engr. LiverpoolSquare Rain Gauge, by Lloyd Corp.  
M.O. 1029

|               |         |       |       |       |       |       |       |       |         |       |                 |
|---------------|---------|-------|-------|-------|-------|-------|-------|-------|---------|-------|-----------------|
| Diameter ...  | 10 x 10 |       |       |       |       |       |       |       | 10 x 10 |       | 8"              |
| Height .....  | 1.10    |       |       |       |       |       |       |       | 1       |       | 1" 0"           |
| Altitude ...  | 84      |       |       |       |       | 82    |       |       | 81-4"   |       |                 |
| Year .....    | 1920    | 1921  | 1922  | 1923  | 1924  | 1925  | 1926  | 1927  | 1928    | 1929  | 1920-<br>MEANS. |
| January ...   | 4.32    | 4.34  | 4.29  | 2.10  | 3.10  | 1.88  | 3.08  | 4.42  | 6.21    | 2.16  |                 |
| February ...  | 3.19    | 2.7   | 3.33  | 4.63  | .89   | 5.73  | 3.52  | 1.73  | 3.31    | 1.20  |                 |
| March .....   | 4.42    | 1.95  | 2.31  | 1.53  | 1.03  | .96   | 1.29  | 3.50  | 2.09    | .60   |                 |
| April .....   | 5.08    | 1.49  | 2.60  | 3.48  | 2.22  | 2.36  | .98   | 2.44  | 1.23    | 1.32  |                 |
| May .....     | 4.69    | 2.52  | 1.46  | 4.05  | 6.90  | 3.80  | 4.29  | 1.52  | 1.50    | 1.91  |                 |
| June .....    | 2.87    | .53   | 2.75  | .70   | 3.20  | .09   | 1.74  | 5.31  | 6.58    | 1.72  |                 |
| July .....    | 6.63    | 2.02  | 4.96  | 3.86  | 6.14  | 3.44  | 4.34  | 4.06  | 2.86    | 3.22  |                 |
| August .....  | 2.36    | 5.30  | 4.13  | 5.02  | 6.39  | 5.05  | 3.86  | 6.09  | 4.00    | 4.00  |                 |
| September...  | 3.59    | 1.28  | 3.95  | 4.41  | 4.67  | 5.63  | 4.22  | 8.04  | 1.91    | 1.91  |                 |
| October ..... | 1.76    | 3.20  | .61   | 5.75  | 4.72  | 5.20  | 5.04  | 3.37  | 4.42    | 5.52  |                 |
| November...   | .69     | 2.50  | 2.03  | 4.31  | 2.49  | 3.13  | 5.03  | 5.36  | 5.65    | 6.12  |                 |
| December...   | 2.74    | 4.41  | 5.22  | 5.40  | 3.79  | 3.02  | 1.37  | 1.91  | 2.67    | 5.69  |                 |
| TOTALS...     | 42.34   | 29.81 | 37.64 | 45.22 | 45.54 | 39.99 | 38.81 | 47.80 | 42.43   | 35.67 |                 |

1920 Iron railing 5 ft high 6 ft enclosing Gauge

1929 Iron railing 4 ft high 5 ft distant from gauge, enclosing rectangular area.

Records 1911-1929 generally too big  
See Memo 27 Feb 815111/28

New gauge on new site in 1930  
See next sheet.

Rejected from Survey, 6 ft high

Position of Station on reduced Ordnance Survey  
Map, 2 miles to 1 inch Series

No. of Map

Reference No.

| Date. | Lat. N.     | Long.      | Railway Station.           | Distance from gauge | Direction | Parish Church. | Distance from gauge | Direction |
|-------|-------------|------------|----------------------------|---------------------|-----------|----------------|---------------------|-----------|
| 1920  | 53° 21' 35" | 2° 54' 35" | Garsington<br>Grinsteadale | 1200 ft             | S         | St. Marys      | 800<br>840 ft       | S         |



RAINFALL at CARSTONCounty of LANCASHIREObserver LIVERPOOL CORPORATIONRiver Basin 2 - LOWER MERSEYT. Molgrew  
J. Rawlinson (CITY ENGINEER)MO Rain Gauge, by

|           |      |      |      |      |      |      |      |      |      |      |      |        |
|-----------|------|------|------|------|------|------|------|------|------|------|------|--------|
| Diameter  |      |      |      |      |      |      |      |      |      |      |      | 6"     |
| Height    |      |      |      |      |      |      |      |      |      |      |      | 1' 0"  |
| Altitude  |      |      |      |      |      |      |      |      |      |      |      | 81' 4" |
| Year      | 1920 | 1921 | 1922 | 1923 | 1924 | 1925 | 1926 | 1927 | 1928 | 1929 | 1930 |        |
| January   |      |      |      |      |      |      |      |      |      |      |      | 4.17   |
| February  |      |      |      |      |      |      |      |      |      |      |      | 2.1    |
| March     |      |      |      |      |      |      |      |      |      |      |      | 3.08   |
| April     |      |      |      |      |      |      |      |      |      |      |      | 2.61   |
| May       |      |      |      |      |      |      |      |      |      |      |      | 1.15   |
| June      |      |      |      |      |      |      |      |      |      |      |      | 1.30   |
| July      |      |      |      |      |      |      |      |      |      |      |      | 5.32   |
| August    |      |      |      |      |      |      |      |      |      |      |      | 4.61   |
| September |      |      |      |      |      |      |      |      |      |      |      | 3.26   |
| October   |      |      |      |      |      |      |      |      |      |      |      | 5.24   |
| November  |      |      |      |      |      |      |      |      |      |      |      | 4.55   |
| December  |      |      |      |      |      |      |      |      |      |      |      | 3.90   |
| TOTALS    |      |      |      |      |      |      |      |      |      |      |      | 39.40  |

1930. Iron Railings 4' high 8' from gauge, enclosing rectangular space.

Position of Station on reduced Ordnance Survey  
Map, 2 miles to 1 inch Series

No. of Map.....

Reference No.....

| Date. | Lat. N. | Long. | Railway Station. | Distance from gauge | Direction from gauge | Parish Church. | Distance from gauge | Direction from gauge |
|-------|---------|-------|------------------|---------------------|----------------------|----------------|---------------------|----------------------|
| 1930  |         |       | CRASSENDALE      | 400 yds             | S.                   | CRASSENDALE    | 800'                | S.                   |

RAINFALL at Garston, near G.County of South LancashireObserver Liverpool Corp.River Basin Mersey - lower - 2Type of Gauge M.O.Annual Average 30.7 in. based on        years obs. w<sup>h</sup> weighted. Period of average 1881-1915 Ref. S  
(see S. Lines. Summary, separately worked) 815 11/28.

|           |       |       |       |       |       |       |       |       |       |       |       |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Diameter  | 5.    |       |       |       |       |       |       |       |       |       |       |
| Height    | 1.0.  |       |       |       |       |       |       |       |       |       |       |
| Altitude  | 81.   |       |       |       |       |       |       |       |       |       |       |
| Year      | 1931  | 1932  | 1933  | 1934  | 1935  | 1936  | 1937  | 1938  | 1939  | 1940  | Means |
| January   | 3.44  | 2.90  | 1.33  | 2.30  | 1.68  | 4.64  | 2.10  | 3.47  | 4.58  | 2.42  |       |
| February  | 2.83  | .06   | 3.20  | .36   | 2.65  | 2.10  | 5.06  | .74   | 2.47  | 2.12  |       |
| March     | .29   | 2.15  | 2.14  | 1.67  | 1.18  | 1.75  | 1.32  | 1.01  | 1.78  | 2.47  |       |
| April     | 2.70  | 3.04  | .94   | 2.23  | 2.64  | 1.29  | 3.40  | .28   | 2.03  | 1.99  |       |
| May       | 2.32  | 4.57  | 1.62  | 1.89  | 1.64  | .83   | 2.61  | 2.69  | 1.44  | 2.36  |       |
| June      | 4.64  | .38   | 1.92  | 1.50  | 3.92  | 3.67  | 1.25  | 2.45  | 2.76  | 1.09  |       |
| July      | 2.56  | 5.32  | 1.63  | 1.01  | 1.39  | 3.10  | 2.02  | 3.12  | 5.04  | 2.94  |       |
| August    | 6.62  | 1.61  | 1.90  | 3.66  | 2.07  | 1.76  | 1.95  | 5.30  | 1.42  | .94   |       |
| September | 3.56  | 3.33  | .86   | 2.30  | 5.10  | 3.42  | 1.73  | 1.60  | .82   | 3.32  |       |
| October   | 1.75  | 6.86  | 3.93  | 5.15  | 4.49  | 1.98  | 2.48  | 4.12  | 2.24  | 3.51  |       |
| November  | 4.20  | 3.22  | 1.56  | 2.01  | 3.89  | 3.19  | 1.96  | 3.67  | 4.48  | 5.78  |       |
| December  | 1.72  | 1.47  | .73   | 4.63  | 3.89  | 3.48  | 3.28  | 3.37  | 2.40  | 2.37  |       |
| TOTALS    | 36.63 | 34.95 | 21.76 | 28.71 | 34.54 | 31.21 | 29.16 | 31.82 | 31.46 | 31.31 |       |
|           | D.    | D.    | D.    | D.    | D.    | D.    | D.    | D.    | D.    | D.    |       |

Railings 4' high 8' all round.

1937 Wood fence 3' high 8' enclosing rest space.

Position of Station on reduced Ordnance Survey Map, }  
1 mile to 1 inch Series

No. of Map.....

Reference No.....

| Date. | Lat. N. | Long. | Railway Station. | Distance from gauge | Direction from gauge | Parish Church.        | Distance from gauge | Direction from gauge |
|-------|---------|-------|------------------|---------------------|----------------------|-----------------------|---------------------|----------------------|
| 1931  |         |       | Grassendale      | 400 yds.            | S.                   | S <sup>t</sup> Mary's | 500'                | S.                   |

Station No. 2810RAINFALL at Garston Reservoir

41

County of South LancashireObserver Liverpool Corps.River Basin Mersey - Lower - 2Type of Gauge M.O.Annual Average 30.7 in. based on ..... years obs. un-weighted. Period of average 1881-1915 Ref. S.File No. 815111/23

| Diameter  | 5"    |       |       |       | 567  | 413   |       |       |       |       |       |  |
|-----------|-------|-------|-------|-------|------|-------|-------|-------|-------|-------|-------|--|
| Height    | 1'0"  |       |       |       |      | 1     |       |       |       |       |       |  |
| Altitude  | 81'   |       |       |       |      | 41    |       |       | 81-3' |       |       |  |
| Year      | 1941  | 1942  | 1943  | 1944  | 1945 | 1946  | 1947  | 1948  | 1949  | 1950  | Means |  |
| January   | 2.114 | 3.26  | 3.58  | 3.83  | 1.78 | 3.46  | 2.15  | 6.75  | 2.86  | 1.11  |       |  |
| February  | 3.26  | 1.27  | 1.67  | 1.27  | 2.56 | 4.95  | .92   | 1.87  | 1.40  | 4.49  |       |  |
| March     | 2.81  | 2.10  | .95   | 1.20  | .97  | 1.03  | 4.62  | .85   | 2.37  | 1.10  |       |  |
| April     | 1.48  | 1.21  | 1.23  | 1.39  | 1.61 | .85   | 2.40  | 1.36  | 2.67  | 3.40  |       |  |
| May       | 1.78  | 3.20  | 2.26  | 2.03  | 3.01 | 1.65  | 3.97  | 1.08  | 3.15  | .95   |       |  |
| June      | .41   | .35   | 2.24  | 2.21  | 2.60 | 3.86  | 2.31  | 3.84  | .86   | 2.15  |       |  |
| July      | 2.45  | 2.89  | 3.87  | 2.19  | 2.24 | 3.88  | 3.31  | 3.19  | 3.13  | 2.12  |       |  |
| August    | 4.88  | 2.40  | 3.35  | 3.42  | 2.28 | 4.65  | .37   | 2.98  | 1.74  | 4.83  |       |  |
| September | .80   | 1.78  | 3.88  | 5.41  |      | 3.85  | 2.69  | 1.66  | 1.06  | 5.21  |       |  |
| October   | 3.46  | 2.80  | 4.04  | 3.99  |      | .73   | 1.10  | 2.29  | 3.40  | 2.26  |       |  |
| November  | 2.58  | .74   | 2.56  | 5.08  |      | 4.96  | 3.60  | 2.00  | 4.86  | 5.01  |       |  |
| December  | 1.06  | 3.20  | 1.43  | 2.73  |      | 3.43  | 2.76  | 2.93  | 4.53  | 3.37  |       |  |
| TOTALS    | 27.11 | 25.20 | 31.06 | 34.81 | inc  | 37.30 | 30.20 | 30.80 | 52.03 | 36.00 |       |  |
|           | D     | D     | D     | D     | D    | D     | D     | D     | D     | D     |       |  |

1941 Wood fence 3' high 8' enclosing rect. space.

1945 n.p. inc. Gauge out of order.

Position of Station on reduced Ordnance Survey Map,  
1 mile to 1 inch Series

No. of Map.....

Reference No.....

| Date. | Lat. N. | Long. | Railway Station.           | Distance from gauge | Direction from gauge | Parish Church. | Distance from gauge. | Direction from gauge. |
|-------|---------|-------|----------------------------|---------------------|----------------------|----------------|----------------------|-----------------------|
| 1941  |         |       | Bressington & Grassendale. | 400 y.              | S.                   | St. Marys.     | 300 y.               | S.                    |

BRITISH RAINFALL ORGANIZATION



Station No. 2810

51

RAINFALL at Garston DepotCounty of S. LancashireObserver Liverpool Corpn.River Basin Mersey - Lower 2Type of Gauge the B.A.Annual Average 30.7 in. based on 10 years obs. un-weighted. Period of average 1881-1915 Ref. S  
" " 31.56 " " 10 " " 1948-57 " " 1916-50 File No. 815/11/28

| Diameter ...  | 8"    | 5"    |       | 567   | 413   |       |       |       |       |       |       |
|---------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Height ...    | 1' 0" |       |       |       | 1     |       |       |       |       |       |       |
| Altitude ...  | 81'   |       |       |       | 51    |       |       |       |       |       |       |
| Year ...      | 1951  | 1952  | 1953  | 1954  | 1955  | 1956  | 1957  | 1958  | 1959  | 1960  | Means |
| January ...   | 2.22  | 3.27  | 1.22  | 2.85  | 1.90  | 3.62  | 1.44  | 2.35  | 3.13  | 4.96  |       |
| February ...  | 2.39  | 1.24  | 1.14  | 2.41  | 1.79  | .36   | 1.86  | 4.43  | .26   | 1.68  |       |
| March ...     | 4.73  | 1.69  | 1.04  | 2.03  | 1.73  | 1.41  | 2.17  | 1.39  | 1.34  | 1.41  |       |
| April ...     | 1.35  | 1.83  | 2.17  | .88   | 1.76  | 2.60  | .53   | .88   | 2.94  | 1.28  |       |
| May ...       | 2.93  | 1.75  | 2.09  | 2.42  | 2.48  | .99   | 1.30  | 2.95  | 2.03  | 1.96  |       |
| June ...      | 1.28  | 1.98  | 1.44  | 3.20  | 3.23  | 2.69  | 1.89  | 4.37  | 2.55  | 1.28  |       |
| July ...      | 1.53  | 1.76  | 3.33  | 3.89  | .66   | 4.79  | 3.87  | 4.02  | 3.99  | 3.92  |       |
| August ...    | 4.18  | 1.69  | 3.24  | 3.64  | .98   | 8.97  | 3.04  | 3.40  | 1.15  | 4.53  |       |
| September ... | 2.54  | 4.03  | 2.77  | 2.60  | 2.53  | 2.43  | 6.51  | 4.33  | .17   | 3.60  |       |
| October ...   | .92   | 3.94  | 1.45  | 6.23  | 2.00  | 3.05  | 2.73  | 3.19  | 2.98  | 3.81  |       |
| November ...  | 4.05  | 2.82  | 2.55  | 5.02  | 1.56  | 1.58  | 1.23  | 1.88  | 4.22  | 5.26  |       |
| December ...  | 4.47  | 3.33  | 1.39  | 3.83  | 3.54  | 2.01  | 2.99  | 3.86  | 4.51  | 3.00  |       |
| TOTALS ...    | 35.59 | 29.32 | 23.83 | 39.00 | 24.16 | 34.50 | 29.56 | 37.05 | 29.27 | 36.69 |       |
|               | D     | D     | D     | D     | D     | D     | D     | N.P.  | N.P.  | N.P.  |       |

1951 Flood Fence 3' at 8' enclosing Rect. Sheds.  
 July III 1957 J.B.O.K. J. moved to roof site some time after 1944 and still there. Rain 18" above roof which is 10' above ground level. Wooden platform 15' 25" NW. of edge of flat roof with side running NE-SW with drop of 10' 6" SE. Area marked SE-N-W. J. to be re-sited at P.G.A. 382853. Uncont. f.t. measure.  
 1958 up site unorthodox  
 1959:- up site unorthodox  
 1960:-

Position of Station on Ordnance One-Inch map: No. of Map 100 National Grid Reference 33/395253

| Date. | Lat. N. | Long. | Railway Station.       | Distance from gauge | Direction from gauge | Parish Church. | Distance from gauge. | Direction from gauge. |
|-------|---------|-------|------------------------|---------------------|----------------------|----------------|----------------------|-----------------------|
| 1951  |         |       | Gressington & Garsdale | 400                 | S.                   | St. Marys      | 200                  | S.                    |

Summary of Earlier Records from 10-year Sheets:

| Record D, or d | 1910-19 | 1920-30 | 1931-40 | 1941-50      | 1951-60 |
|----------------|---------|---------|---------|--------------|---------|
|                |         | C1930   | ✓       | 41-44, 46-50 |         |
|                |         | -       | ✓       | ✓            |         |